The Profit Curve

INTEREST RATES AND U.S. BANK PROFITABILITY, 2001–2023

Karan Baylav

DATA ANALYTICS FOR ECONOMIC
ANALYSIS WITH PYTHON

Goal

1. Analyze how the Federal Funds Rate influences U.S. bank profitability, indicated by : Return on Assets (ROA), Return on Equity (ROE), and Net Interest Margin (NIM).

2. Conduct exploratory data analysis (EDA) to examine profitability trends, rate cycles, and correlations before formal modeling.

3. Test for a nonlinear (inverted-U) relationship between interest rates and bank profitability, identify the peak profitability point using econometric and ML methods.

Data Description

Data Sources

- Kaggle: U.S. Banks Financial Quarterly Ratios (2001–2023)
  - Covers profitability and balance-sheet ratios for thousands of banks.
- FRED: Federal Funds Rate (FEDFUNDS)
  - U.S. short-term policy rate, in quarterly frequency.

Cleaning Steps

- Standardized to quarterly format
- Converted numeric variables to consistent types
- Removed missing values and extreme outliers (IQR trimming)
- Final balanced panel: ~640k bank–quarter observations

Merged Dataset

- Matched on quarter, forming a panel of shape (12, ~640k)
- Key variables used in analysis:
  - Profitability: ROAQ, ROEQ, NIMQR
  - Policy: FEDFUNDS, FEDFUNDS_CHANGE
  - Controls: equity_ratio, loan_ratio, deposit_ratio, funding_ratio

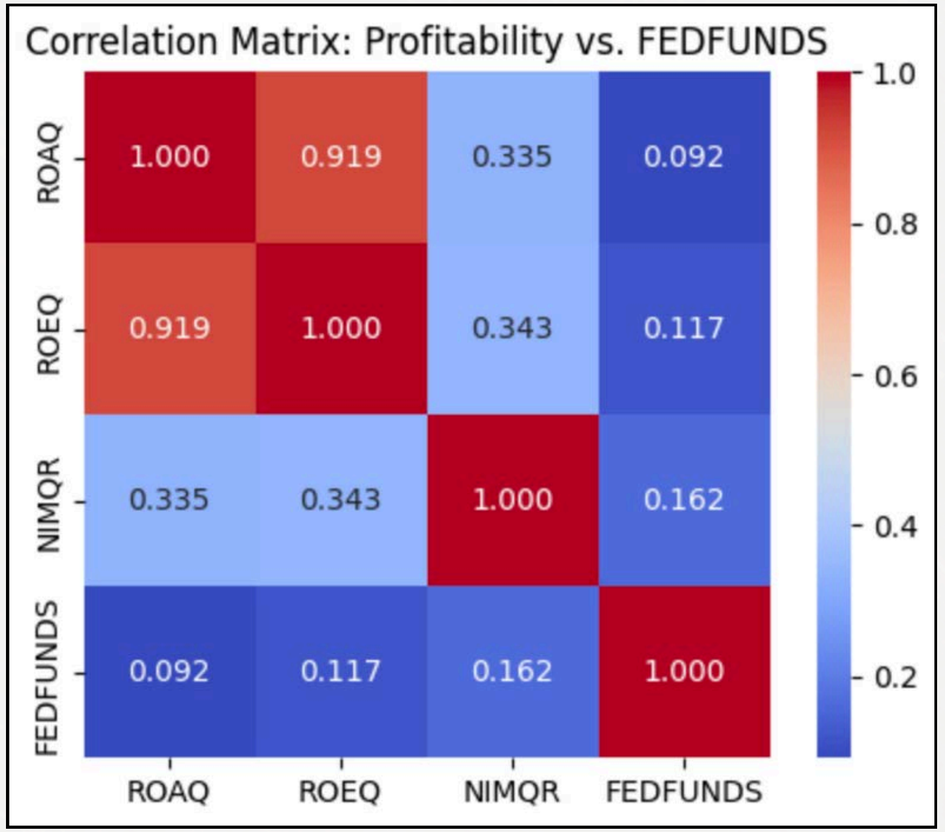
	CERT	QUARTER	ROAQ	ROEQ	NIMQR	FEDFUNDS	FEDFUNDS_CHANGE	equity_ratio	loan_ratio	deposit_ratio	funding_ratio	deposit_funding_ratio
0	10004	2023Q1	0.0051	0.0597	2.819395	4.52	0.87	9.350400	82.225829	82.981486	90.649600	0.915409
1	10011	2023Q1	0.0085	0.1686	3.160201	4.52	0.87	5.260864	67.435827	79.674906	94.739136	0.840993
2	10012	2023Q1	0.0023	0.0228	2.630143	4.52	0.87	10.329880	59.174107	86.055159	89.670120	0.959686

Summary Statistics & Correlations

Key Insights

- Profitability ratios are positive on average, stable over time.
- FEDFUNDS varies widely, reflecting full monetary cycles.
- Moderate positive correlation between rates and profits before controlling for bank effects.
- Data distribution clean after IQR trimming — suitable for regression analysis.

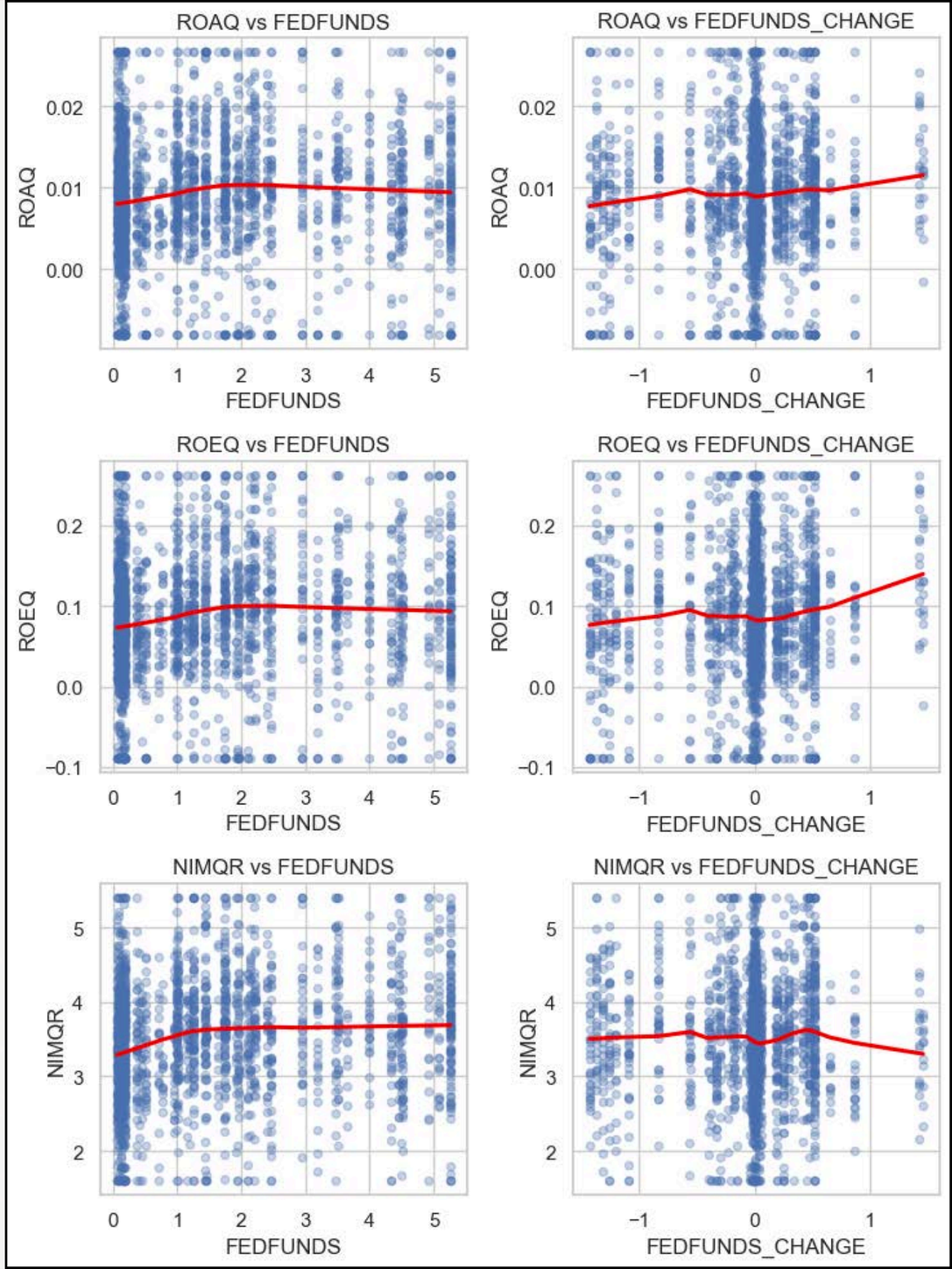
	mean	std	min	max	range
ROAQ	0.009	0.008	-0.008	0.027	0.035
ROEQ	0.086	0.078	-0.089	0.263	0.352
NIMQR	3.507	0.784	1.588	5.420	3.832
FEDFUNDS	1.549	1.646	0.060	5.260	5.200



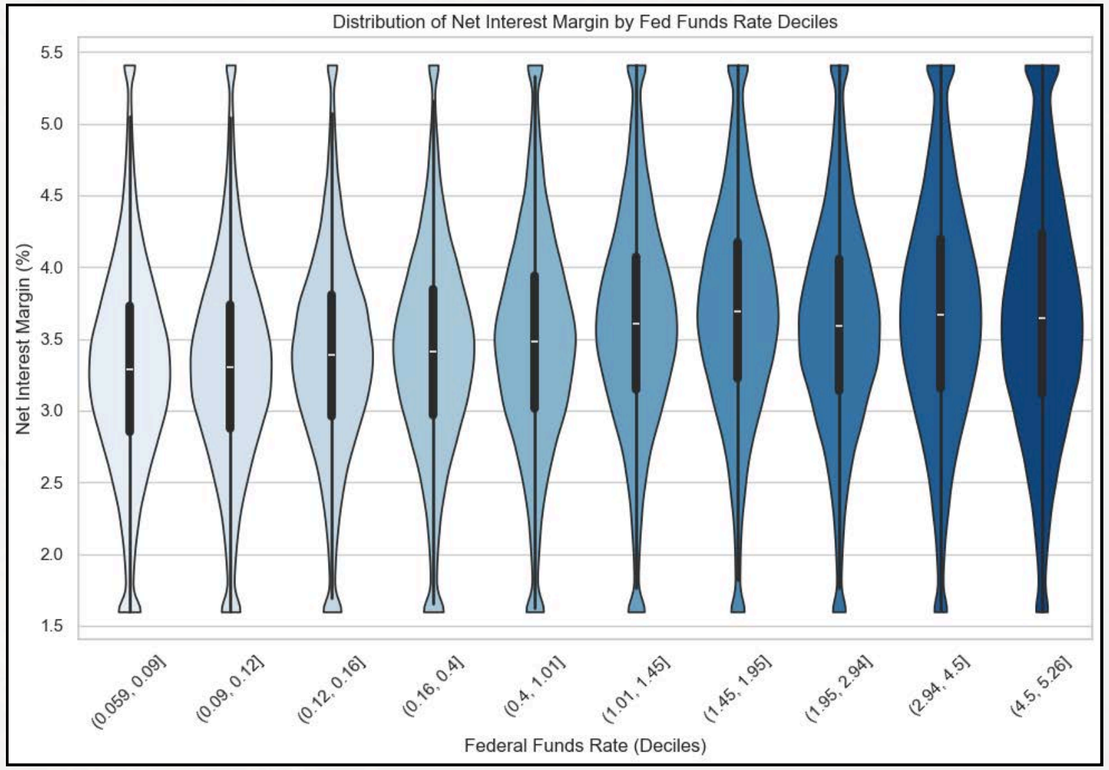
Scatterplots & Time Trends

- Profitability increases with moderate interest rate hikes, then levels off — inverted-U pattern.
- Short-term changes (ΔFEDFUNDS) show small positive effects on ROA and ROE.
- NIM responds most strongly to tightening cycles, as lending margins widen.
- Greater dispersion at higher rates → banks differ in ability to adjust funding structures.

Moderate rate hikes boost margins via lending spreads, while high rates reduce profitability as funding costs rise—justifying the quadratic term in later regressions.



Profitability Distributions Across Rate Regimes



- Each violin shows the distribution of Net Interest Margins (NIM) across banks for a given interest-rate range.
- Low-rate regimes (left side) show compressed, low profitability.
- Mid-rate ranges (≈2–4%) display higher median NIM and wider dispersion, indicating stronger and more varied bank performance.
- At higher rates, NIM plateaus, suggesting limited additional gains.
- The pattern illustrates a nonlinear, inverted-U relationship between interest rates and bank profitability.

Quadratic Regression Project Details

Empirical Strategy

Estimate a quadratic panel model with Bank Fixed Effects to isolate within-bank variation over time. Exclude time fixed effects, since the Federal Funds Rate is identical across banks each quarter and would be absorbed.

Include balance-sheet controls (equity, loan, deposit, and funding ratios) to capture structural differences across banks.
Use Gradient Boosting + SHAP as a model-free validation of the inverted-U pattern.

Regression Model Design

$$Profitability_{it} = \beta_0 + \beta_1 \cdot FEDFUNDS_t + \beta_2 \cdot FEDFUNDS_t^2 + \gamma \cdot X_{it} + \mu_i + \varepsilon_{it}$$

Bank Fixed Effects capture time-invariant differences; time FE excluded since FEDFUNDS is identical across banks. The model includes structural controls and uses bank-clustered standard errors for robustness.

Quadratic OLS with Controls

$$PROFITABILITY_{it} = B_0 + B_1 \cdot FEDFUNDS_t + B_2 \cdot FEDFUNDS_t^2 + \Gamma \cdot X_{it} + E_{it}$$

Variable	Coefficient	ROAQ	ROEQ	NIMQR
FEDFUNDS	β_1	+0.0018***	+0.0197***	+0.2101***
FEDFUNDS²	β_2	-0.0003***	-0.0029***	-0.0289***
Peak Rate	—	**3.3%**	**3.4%**	**3.6%**

A 1pp rise in the Fed Funds Rate (from 2% to 3%) is associated with a ~0.21pp increase in NIM, holding structure constant.

- **Heteroskedasticity**: BP test ($p<0.001$) → robust SEs
- **Serial correlation**: DW ≈ 1.7 → mild, corrected by clustering
- **Specification**: RESET $p<0.001$ → supports quadratic term

FEDFUNDS and FEDFUNDS² remain significant with expected signs; profitability peaks around 3.5%, and the inverted-U pattern stays robust after adding structural controls.

Quadratic OLS with Controls & Bank FE

$$PROFITABILITY_{it} = B_0 + B_1 \cdot FEDFUNDS_t + B_2 \cdot FEDFUNDS_t^2 + \Gamma \cdot X_{it} + M_i + E_{it}$$

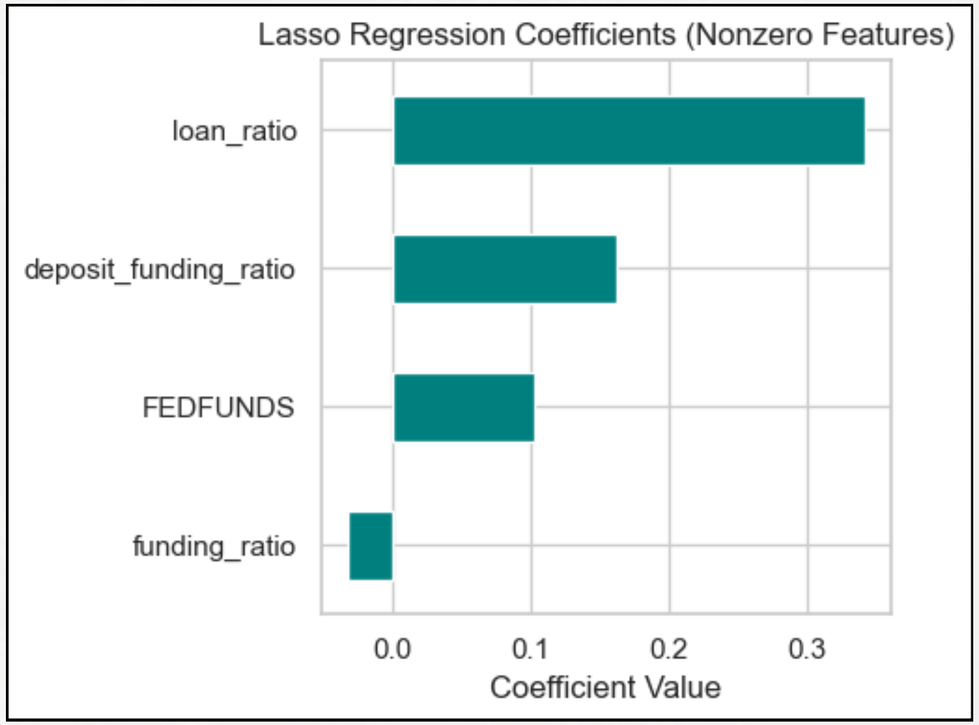
Variable	ROAQ	ROEQ	NIMQR
FEDFUNDS (β_1)	+0.0015***	+0.0176***	+0.1761***
FEDFUNDS² (β_2)	-0.0002***	-0.0025***	-0.0243***
Equity Ratio	-0.0106***	-0.0974***	-0.3341**
Loan Ratio	+0.00009***	+0.0007***	+0.0242***
Deposit Ratio	+0.0004***	+0.0027***	+0.0027***
Funding Ratio	-0.0112***	-0.0959***	-0.3747**
Deposit/Funding Ratio	-0.0355***	-0.2212***	+0.5173***
Peak Profitability (Rate)	**3.48%**	**3.57%**	**3.62%**

A 1pp rise in the Fed Funds Rate (from 2% to 3%) is associated with a ~0.18pp increase in NIM, holding structure constant.

All key coefficients are highly significant; FEDFUNDS positive and FEDFUNDS² negative confirm an inverted-U, with profitability peaking near 3.5%. Bank FE control for fixed traits, and time FE are excluded since FEDFUNDS is identical across banks.

Lasso Regression (Feature Selection)

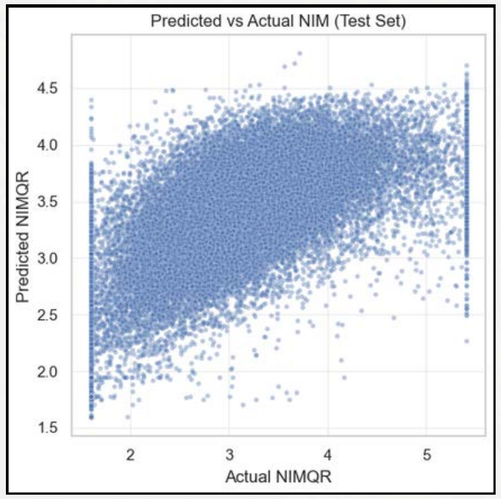
target: NIMQR

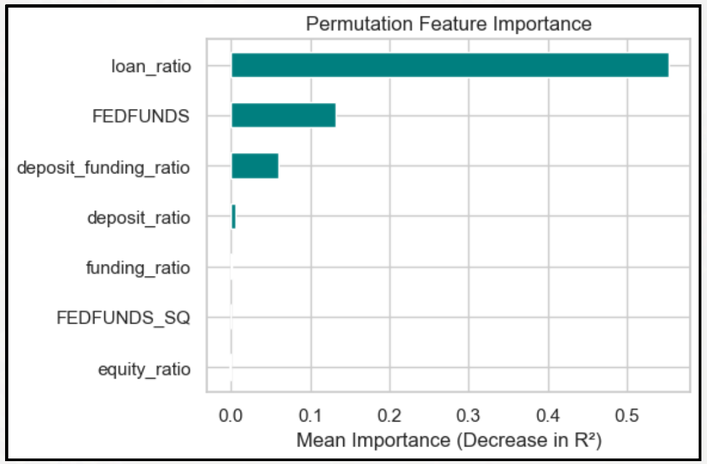


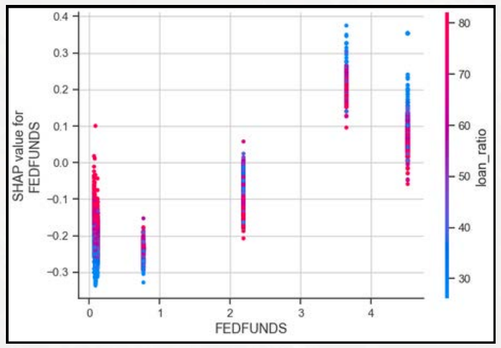
Cross-validation automatically selects the optimal regularization penalty (α), balancing model simplicity and predictive power.

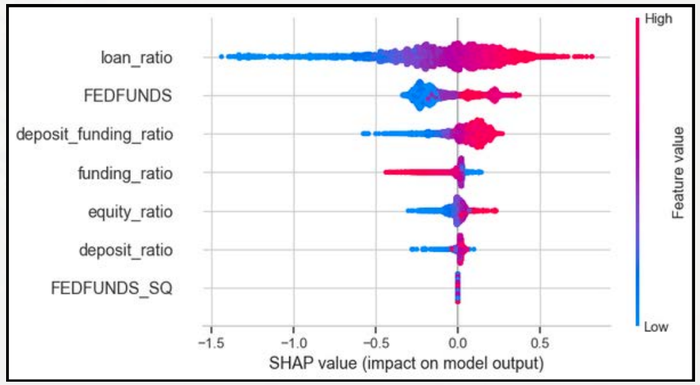
- The Lasso regression applies L1 regularization, penalizing weaker predictors and keeping the most influential drivers of profitability.
- loan_ratio and deposit_funding_ratio dominate, emphasizing the importance of a bank's structural composition in shaping outcomes.
- FEDFUNDS remains significant under regularization, confirming that monetary policy effects persist once structural factors are accounted for.
- By shrinking minor coefficients toward zero, the model highlights key variables while reducing noise and overfitting risk.
- Overall, the Lasso results reinforce the robustness of the econometric findings, showing that both bank structure and policy rates are central to explaining profitability.

Gradient Boosting + SHAP









Target (y): NIMQR

- The Gradient Boosting model predicts NIM with good accuracy (RMSE ≈ 0.65), capturing cross-bank profitability patterns.
- Feature importance shows that structural factors—especially loan activity and deposit funding—explain most variation in NIM, while the policy rate (FEDFUNDS) remains a key driver.
- SHAP analysis confirms a nonlinear relationship between the Federal Funds Rate and profitability—margins rise up to ~3–4 % and then flatten.
- The model replicates the inverted-U pattern from the econometric analysis, validating it without assuming a functional form.
- Together, boosting and SHAP show how bank structure and monetary policy interact to shape profitability.

Policy Implications

- The analysis identifies an optimal policy rate range of roughly 3–4 %, where bank profitability is highest without increasing financial stress.
- Moderate interest rates enhance lending margins and stability, supporting sustainable credit supply across institutions.
- Prolonged high-rate environments compress margins and could reduce risk appetite, particularly among smaller or less diversified banks.
- Low-rate periods narrow spreads and incentivize excessive risk-taking, underscoring the trade-off faced by monetary authorities.
- Policymakers should balance profitability, credit growth, and stability objectives, using rate adjustments that maintain efficient intermediation while avoiding systemic strain.
- Integrating econometric and explainable ML approaches offers a practical framework for monitoring profitability risks under different monetary policy scenarios.



Sources & Acknowledgments

Data Sources

- Kaggle: U.S. Banks Financial Quarterly Ratios (2001–2023) — provides profitability and balance-sheet indicators for ~6,000 U.S. banks.
- FRED (Federal Reserve Economic Data): Federal Funds Rate (FEDFUNDS) — quarterly policy rate data merged by reporting period.

Methods & Tools

- Econometrics: Pooled OLS, Quadratic OLS, and Bank Fixed Effects regressions using statsmodels and linearmodels.
- Machine Learning: Gradient Boosting (scikit-learn) with SHAP explainability, and Lasso regression for feature selection.
- Data Processing & Visualization: pandas, numpy, matplotlib, and seaborn (Python 3.12.2).

Acknowledgments & Limitations

- Data coverage: limited to reporting institutions; smaller or non-surviving banks may be underrepresented.
- Econometric limits: FEDFUNDS is common across banks, so time FE were excluded; results describe associations, not causality.
- Modeling limits: ML confirms robustness but does not account for macroeconomic shocks or structural policy changes.
- All code, cleaning, and analysis performed independently in Python 3.12.2, with syntax verification and formatting assisted by AI-based tools.

Appendix:
Pyhton Code of the Models

Quadratic OLS with Controls

```
# === 12) Quadratic OLS with Controls (Clustered by Bank) ===

controls = "equity_ratio + loan_ratio + deposit_ratio + funding_ratio + deposit_funding_ratio"

for y in targets:
    # --- Estimate model with controls and clustered SEs ---
    model_ctrl = ols(
        f"{y} ~ FEDFUNDS + I(FEDFUNDS**2) + {controls}",
        data=df
    ).fit(
        cov_type='cluster',
        cov_kwds={'groups': df['CERT']}
    )

    # --- Output summary ---
    print(f"\n=== Quadratic OLS with Controls (Clustered by Bank): {y} ===")
    print(model_ctrl.summary().tables[1])

    # --- Compute turning point (peak profitability) ---
    b1 = model_ctrl.params['FEDFUNDS']
    b2 = model_ctrl.params['I(FEDFUNDS ** 2)']
    x_star = -b1 / (2 * b2)

    print(f"Peak profitability (with controls) for {y}: {x_star:.2f}% FEDFUNDS\n")
```

Quadratic OLS with Controls & Bank FE

```python
# === 18) Fixed Effects (Bank only) Quadratic Model ===

from linearmodels.panel import PanelOLS

targets = ['ROAQ', 'ROEQ', 'NIMQR']

for y in targets:
    yv = df_panel[y]
    X = df_panel[['FEDFUNDS', 'FEDFUNDS_SQ'] + controls]

    mod = PanelOLS(yv, X, entity_effects=True, drop_absorbed=True)
    res = mod.fit(cov_type='clustered', cluster_entity=True)

    b1 = res.params.get('FEDFUNDS', np.nan)
    b2 = res.params.get('FEDFUNDS_SQ', np.nan)
    x_star = (-b1 / (2 * b2)) if pd.notna(b1) and pd.notna(b2) else np.nan

    print(f"\n=== Bank Fixed Effects Only — {y} ===")
    print(res.summary.tables[1])
    print(f"Peak profitability (Bank FE): {x_star:.2f}% FEDFUNDS")
```

Lasso Regression

```
# === Lasso Regression for Robustness Check ===

# Prepare features and target
features = [
    'FEDFUNDS', 'FEDFUNDS_SQ', 'equity_ratio', 'loan_ratio',
    'deposit_ratio', 'funding_ratio', 'deposit_funding_ratio'
]
X = df[features].dropna()
y = df.loc[X.index, 'NIMQR']

# Build and fit Lasso with cross-validation
lasso = make_pipeline(StandardScaler(), LassoCV(cv=5, random_state=42))
lasso.fit(X, y)

# Extract nonzero coefficients
coefs = pd.Series(lasso.named_steps['lassocv'].coef_, index=X.columns)
coefs = coefs[coefs != 0].sort_values(key=abs, ascending=True)

# Plot
plt.figure(figsize=(5,4))
coefs.plot(kind='barh', color='teal')
plt.title('Lasso Regression Coefficients (Nonzero Features)')
plt.xlabel('Coefficient Value')
plt.tight_layout()
plt.show()

print("Selected features:", list(coefs.index))
```

Gradient Boosting + SHAP

```python
# === 1) Imports ===
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.ensemble import HistGradientBoostingRegressor
from sklearn.metrics import mean_squared_error
from sklearn.inspection import permutation_importance

import shap

pd.set_option("display.max_columns", 200)
pd.set_option("display.width", 120)
sns.set(context="notebook", style="whitegrid")

# === 2) Data ===
# df must already be loaded with columns: QUARTER, NIMQR, FEDFUNDS, equity_ratio, loan_ratio, deposit_ratio, funding_ratio, deposit_funding_ratio
feat = ["FEDFUNDS", "equity_ratio", "loan_ratio", "deposit_ratio", "funding_ratio", "deposit_funding_ratio"]
df["FEDFUNDS_SQ"] = df["FEDFUNDS"] ** 2
X = df[feat + ["FEDFUNDS_SQ"]].copy()
y = df["NIMQR"].copy()

mask = X.notnull().all(1) & y.notnull() & df["QUARTER"].notnull()
X, y, q = X[mask], y[mask], pd.PeriodIndex(df.loc[mask, "QUARTER"], freq="Q")

order = np.argsort(q.astype(str).values)
X, y, q = X.iloc[order], y.iloc[order], q[order]

cut = q.max() - 8  # last 8 quarters = test
train_idx = q <= cut
test_idx  = q >  cut

X_tr, y_tr = X[train_idx], y[train_idx]
X_te, y_te = X[test_idx],  y[test_idx]

print("Train/Test sizes:", X_tr.shape, X_te.shape)
```

Gradient Boosting + SHAP

```python
# === 3) Model: HistGradientBoostingRegressor ===
model = HistGradientBoostingRegressor(
    max_iter=600,
    learning_rate=0.03,
    max_depth=4,
    random_state=42
)
model.fit(X_tr, y_tr)

# === 4) Eval ===
pred = model.predict(X_te)
mse = mean_squared_error(y_te, pred)
rmse = np.sqrt(mse)
print(f"Test RMSE: {rmse:.4f}")

plt.figure(figsize=(5,5))
sns.scatterplot(x=y_te, y=pred, alpha=0.4, s=12)
plt.xlabel("Actual NIMQR")
plt.ylabel("Predicted NIMQR")
plt.title("Predicted vs Actual NIM (Test Set)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

Gradient Boosting + SHAP

```python
# === 5) Permutation Feature Importance ===
perm = permutation_importance(model, X_te, y_te, n_repeats=10, random_state=42, n_jobs=-1)
imp = pd.Series(perm.importances_mean, index=X_te.columns).sort_values(ascending=True)

plt.figure(figsize=(6,4))
imp.plot(kind="barh", color="teal")
plt.title("Permutation Feature Importance")
plt.xlabel("Mean Importance (Decrease in R²)")
plt.tight_layout()
plt.show()

# === 6) SHAP Explainability ===
SAMPLE = min(4000, len(X_te))
Xs = X_te.sample(SAMPLE, random_state=42)

# model-agnostic explainer for sklearn HGBR
background = X_tr.sample(min(2000, len(X_tr)), random_state=42)
explainer = shap.Explainer(model.predict, background)
shap_values = explainer(Xs)

# SHAP summary (global)
shap.summary_plot(shap_values.values, Xs, feature_names=Xs.columns, show=True)

# SHAP dependence for the policy rate
shap.dependence_plot("FEDFUNDS", shap_values.values, Xs, feature_names=Xs.columns, show=True)
```